

Financial Statements and Company Valuation

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1 Financial Reports and Statements

There are 3 basic financial statements included in reports:

- Balance sheet (B/S)
- Income statement (I/S)
- Cash flow statement (CFS)

These 3 financial statements are found in financial reports of various accounting periods. Under U.S. SEC regulations, annual reports are called 10-k and quarter reports are called 10-Q.

1.1 Income Statement

An income statement (I/S) starts with revenue and ends with earnings per share. A typical I/S contains the following line items (but are in no way limited to them):

- Revenue
- Cost of goods sold (CoGS), cost of sales, cost of contracts: the cost of producing the goods sold by a company
- Gross profit: revenue less CoGS
- Operating expense (opex): recurring cost of maintaining operations such as payrolls, rent, and utilities. Common subitems include:
 - General and administrative expenses (G&A)
 - Marketing and advertizing
 - Research and development expenses (R&D)
- Operating profit: gross profit less operating expense. Some companies add share of gain/loss of joint ventures (JV) to operating profit.
- Unusual, infrequent, non-recurring, or extraordinary items
- Earnings/profit before interests/financing and tax (EBIT): operating profit less unusual items. When there is no unusual items, EBIT and operating profit are often exchangeable.
- Interest income: money earning from savings, papers, certificate of deposits (CDs), or money market funds
- Interest expense: interest of company's loans
- Earnings/profit/net income before taxation (EBT): EBIT plus interest income and less interest expense
- Tax
- Earnings/net income attributable to non-controlling interests (EANCI): the portion of earnings that belong to subsidiaries which the parent company has less than 50% shares in.
- Earnings/net income attributable to common/ordinary shareholders: EBT less earnings of NCI

When we put the items together, we get the formula of net income (to common):

$$\begin{aligned} \text{NI} &= \text{Revenue} \\ &\quad - \text{CoGS} \\ &\quad - \text{Opex} \\ &\quad - \text{Unusual items} \\ &\quad + \text{Interest income} \\ &\quad - \text{Interest expenses} \\ &\quad - \text{Tax} \\ &\quad - \text{EANCI} \end{aligned}$$

Figure 1 shows the income statement of NVIDIA from 2024 Q2.

1.2 Cash Flow Statement

A cash flow statement lists cash inflow and cash outflow from three main categories:

1. Operating cash flow (OCF): cash inflow/outflow related to core operation
2. Investing cash flow: cash inflow/outflow related to investments (short or long term), such as purchases maintenance and disposals of assets, acquisitions, and investments in securities
3. Financing cash flow: cash inflow/outflow related to debts and equity, such as debt issuance and repayment, dividends, and share buybacks

There are two ways to record cash flows:

	Three Months Ended		Six Months Ended	
	July 30, 2023	July 31, 2022	July 30, 2023	July 31, 2022
Revenue	\$ 13,507	\$ 6,704	\$ 20,699	\$ 14,992
Cost of revenue	4,045	3,789	6,589	6,646
Gross profit	9,462	2,915	14,110	8,346
Operating expenses				
Research and development	2,040	1,824	3,916	3,443
Sales, general and administrative	622	592	1,253	1,183
Acquisition termination cost	—	—	—	1,353
Total operating expenses	2,662	2,416	5,169	5,979
Operating income	6,800	499	8,941	2,367
Interest income	187	46	338	64
Interest expense	(65)	(65)	(131)	(132)
Other, net	59	(5)	42	(19)
Other income (expense), net	181	(24)	249	(87)
Income before income tax	6,981	475	9,190	2,280
Income tax expense (benefit)	793	(181)	958	6
Net income	\$ 6,188	\$ 656	\$ 8,232	\$ 2,274
Net income per share:				
Basic	\$ 2.50	\$ 0.26	\$ 3.33	\$ 0.91
Diluted	\$ 2.48	\$ 0.26	\$ 3.30	\$ 0.90
Weighted average shares used in per share computation:				
Basic	2,473	2,495	2,472	2,500
Diluted	2,499	2,516	2,495	2,526

Figure 1: NVIDIA corporation condensed consolidated statement of income in millions [4].

1. Direct method: each inflow and outflow is recorded and presented on the statement
2. Indirect method: net cash flow is worked out by adjusting other financial figures

OCF is mostly recorded using indirect method. As such, it is usually worked out in 2 steps, starting from NI:

1. Adjustments to reconcile NI to OCF: non-cash expenses (determined by the company or in an common accounting sense) are added back since they were deducted from NI previously
2. Changes in operating assets and liabilities, collectively knows as *working capitals* (WC)

Some companies have a separate line stating how much taxes are paid in cash. See the section on taxes below for tax treatment in cash flow statements.

$$\text{OCF} = \text{NI} + \text{Non-cash expenses} - \Delta\text{WC}$$

1.2.1 Non-Cash Expenses

Losses and decrease in values arising from practices of accrual accounting are recorded as non-cash charges/expenses in the income statement and deducted from revenue. Since these are not real cash flows, they are added back to net income to arrive at the correct number of operating cash flow. While companies usually have some discretion in determining what non-cash expenses to put into OCF, common non-cash adjustments include:

- Depreciation and amortization (D&A)
- Stock-based compensation (SBC)
- Deferred income taxes
- Loss on disposal of assets
- Gain/losses on investments in joint ventures or non-affiliates

- Gain/losses in foreign exchanges

Adjustments that are just tomfoolery and should NOT appear in OCF include:

- Dividends received from investments
- “Other” long-term liabilities

Depreciation, amortization, and gain/loss on asset disposal arise from the difference between accrual accounting and cash accounting. When an asset is bought at a price and assumed to be sold at a lower price (*salvage value*), the difference between these two prices is spread cross years (*useful life*). Decrease in the value of tangible assets is called *depreciation* while that of intangible assets is called *amortization*. Since D&A are not real cash costs, they are added back to NI during reconciliation.

When assets are sold at a price different from assumed salvage value, gain/loss occurs and is recognized in income statement. Again, this is not a real cash item, so it is removed from NI.

Let us consider a hypothetical example. A company buys a car for \$100,000 which is depreciated over 5 years. Its salvage value is assumed to be \$50,000. It is sold for \$65,000 at year 6. The change in shareholder’s equity from year 5 to year 6, which amounts to \$5,000, is exactly the number of gain on asset disposal minus depreciation expense.

Year	6	5	4	3	2	1
I/S:						
Depreciation expense	10,000	10,000	10,000	10,000	10,000	0
Gain/(loss) on asset disposal	15,000	0	0	0	0	0
CFS:						
Purchase of assets	0	0	0	0	0	100,000
Proceeds from disposal of assets	65,000	0	0	0	0	0
B/S:						
Accumulated depreciation	0	40,000	30,000	20,000	10,000	0
Net PP&E	0	60,000	70,000	80,000	90,000	100,000
Cash	65,000	0	0	0	0	0

Table 1: D&A and gain/(loss) on sales of assets.

1.2.2 Changes in Working Capitals

When a company places orders for raw material or sells goods to customers, cash transfer does not always occur at the moment of billing. This time difference begets the items of *working capitals* in B/S. In the most narrow sense of working capital, there are three items to consider:

- Accounts payable (A/P): goods and services purchased on credit
- Inventories: raw materials, work-in-progress goods, and finished goods
- Accounts receivable (A/R): the money owed by customers that paid using credit

In accrual accounting, the credits are recognized when they are billed not when the cash transfer is finished, so A/P, even though recognized in I/S as part of CoGS, does not actually mean cash outflow. Similarly, A/R, which is recognized as revenue, does not mean cash inflow either. If there is an increase in A/P, NI decreases, and to arrive at the cash flow figure we add it back. Likewise, an increase in A/R must be deducted from NI to arrive at OCF. Thus, overall, the reconciliation regarding working capital would be to *subtract* the change in NWC.

$$\Delta \text{NWC} = \Delta \text{A/R} + \Delta \text{Inventory} - \Delta \text{A/P}$$

The time span between paying cash to suppliers and receiving cash from customers is called *cash conversion cycle* (CCC). Figure 2 shows components of a CCC. For companies such as Rolls-Royce that manufacture jet engines, CCC could last longer than a year, and its working capital is divided into two categories: *current* WCs – A/R and A/P that can be paid and collected within one year – and non-current WCs that are not expected to be paid or collected in a year.

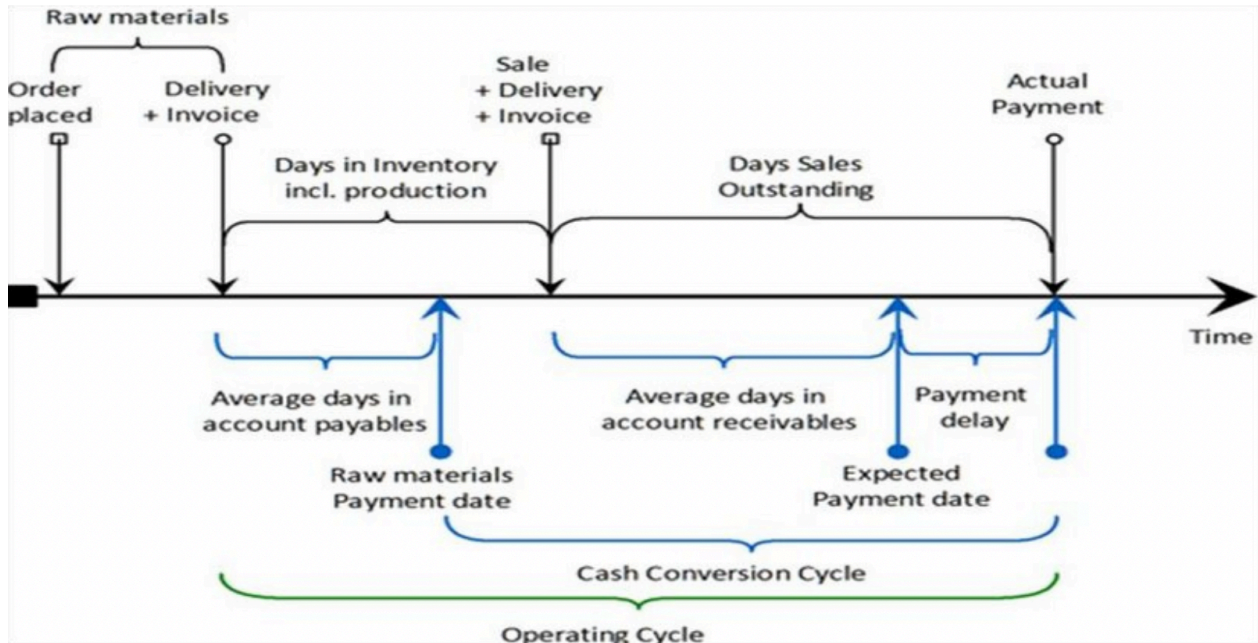


Figure 2: Components of a cash conversion cycle.

Broader kind of working capital This time difference between accrued items and cash transfer can be applied to other items such as payrolls, expenses, prepaid items. As such we expand the definition of net working capital, and in some definitions we even include cash and short-term debts. We end up at least four kinds of working capital:

1. *Net* working capital (NWC)
2. *Non-cash* working capital (NCWC)
3. *Operating* working capital (OWC)
4. *Trade* working capital (TWC)

NWC includes all kinds of current assets – all moneys that can be spent immediately and capital that can be converted to cash in a year – and current liabilities – all expenses that must be paid in short term. NCWC excludes cash or *excess* cash. OWC only includes “operational” current assets and “operational” current liabilities, which is NCWC minus debts (short-term debts and long-term debts due in a short time). TWC is strictly defined as $A/R + Inventory - A/P$.

1.2.3 Taxes

A company does not pay exactly the amount of taxes it should based on its reported pretax income. Companies could defer or over pay taxes, receive tax credits, or report a completely different taxable income figure to IRS to “cheat” IRS. The difference between tax expense (tax based on reported pretax income) and tax payable (aka. current tax, tax actually paid in cash) is thus *deferred tax*.

$$\text{Tax expense}/(\text{benefit}) = \text{Tax payable} + \text{Deferred tax}$$

	NWC	NCWC	OWC	TWC
Current asset	• A/R • Inventory • Prepaid expense • Cash and securities	• A/R • Inventory • Prepaid expense	• A/R • Inventory • Prepaid expense	• A/R • Inventory
Current liability	• A/P • Taxes, rent, payrolls • Short-term debts • Principle part of long-term debts • Long-term debts due soon	• A/P • Taxes, rent, payrolls • Short-term debts • Principle part of long-term debts • Long-term debts due soon	• A/P • Taxes, rent, payrolls	• A/P

Table 2: Different definitions of trading capitals.

Overpaid taxes become *deferred tax assets* (DTA) and underpaid taxes become *deferred tax liabilities* (DTL). DTA and DTL show up in B/S.

When companies reconcile NI to OCF, US companies subtract deferred taxes from NI while companies outside of US add tax expense and then subtract tax payable.

Tax rates can be derived from the tax figures in financial reports, including *effective tax rate* and *cash tax rate*.

$$\text{Effective tax rate} = \frac{\text{Tax expense}}{\text{Pretax income}} \quad (1)$$

$$\text{Cash tax rate} = \frac{\text{Tax payable}}{\text{Pretax income}} \quad (2)$$

In cases where companies deferred taxes too much to the extent that effective tax rate becomes negative in a certain year, average effective tax rate in the past or cash tax rate is a good substitute for valuation.

1.2.4 Interests

Interests received or paid do not appear in CFS of some companies because they are fully accounted for in net income. One could argue that net interest income/expense includes non-cash items such as foreign exchange and gain/loss in derivatives and we must adjust for these by adding net interest income/expense in the reconciliation section and also adding the interests received/paid in cash somewhere else.

Let us consider a hypothetical example with net financing in Table 3. In I/S the number we deduct from operating income as interest cost would be \$(510)m. If we want to adjust for non-cash interests, \$510m would be added back to NI to arrive at OCF, and \$269m and \$(362)m would be in the section of cash flow from financing. By doing so, the net cash flow would have $510 + 269 - 362 = 417$ million dollars more than before, exactly the net loss in FX and derivative.

After the adjustment OCF sees an increase of \$510m. Now, while IFRS allows interest paid to be classified as operating or financing, GAAP *requires* it to be operating, so if we classify it as operating, OCF would instead see an increase of \$779m. In the process we essentially inflated OCF.

Interest receivable from cash and equivalents	269
Net foreign exchange gains	191
Net fair value gains on interest rate swaps	40
Financing income	500
Interest payable	(362)
Net fair value losses on foreign currency contracts	(630)
Net fair value losses on commodity contracts	(18)
Financing cost	(1,010)
Net interest income/(expense)	(510)

Table 3: Net financing of a hypothetical company, with cash portions being interest receivable and interest payable. Numbers are in million dollars.

Take a look at Rolls-Royce's CFS and compare it to BAE System's. Rolls-Royce classifies interest received as operating while BAE System classifies it as investing. Notice that Rolls-Royce does not include net financing cost of £(688)m because it starts its reconciliation from operating income, which is already before interest.

	Rolls-Royce	BAE System
Cash flows from operating activities:		
Operating income	2,906	–
Net income	–	2,041
...		
Net finance costs	–	353
Interest received	269	
...		
Cash flows from investing activities:		
Interest received	–	130
...		
Cash flows from financing activities:		
Interest paid	(200)	(543)
...		

Table 4: Excerpts from FY2024 CFS of Rolls-Royce and BAE System. Numbers are in million pounds

1.2.5 Free Cash Flow and Owner's Earning

There are three metrics to measure the cash that is available after necessary investments for maintenance and growth are made:

1. Free cash flow to the firm (FCFF), also known as *unlevered* free cash flow (UFCF)
2. Free cash flow to equity (FCFE), also known as *levered* free cash flow (LFCF)
3. Owner earnings

FCFE FCFE means the cash that is available after the company buys its assets, pays taxes and pays interests to debt issuers. The shareholders gets the cash the last since they are the bottom of the capital structure. Since OCF already has taxes and change in OWC included, we will start from this figure:

$$\text{FCFE} = \text{OCF} + \text{Net borrowing} - \text{Reinvestment} \quad (3)$$

where

$$\text{Reinvestment} = \text{Capital expenditure} + \text{Purchase of intangibles (+ Acquisition cost)}$$

and

$$\text{Net borrowing} = \text{Debt issued} - \text{Debt repaid}$$

Alternatively, expanding the definition of OCF, we can start FCFE from NI instead,

$$\text{FCFE} = \text{NI} + \text{Non-cash expenses} - \Delta\text{OWC} + \text{Net borrowing} - \text{Reinvestment}$$

Here we see why we would need to use OWC as our definition of working capital instead of a broader NWC. If we include debts in working capital, newly issued debt will increase FCFE, and net borrowing term double counts it.

While it is straightforward that debt repaid decreases FCFE, it could be a little bit hard to understand how debt newly issued increases FCFE. Suppose that we raise new debts to pay dividends (which is common in some oil or utility companies), FCFE must include the cash that is used to pay dividends because it is the cash available for shareholders.

Acquisition cost is included if the company is on an acquiring spree for years. For some, if acquisition is an infrequent item, it can be left out of calculation of cash flows to arrive at a normalized figure, hence the parentheses in the equation.

FCFF FCFF is a hypothetical figure that measures cash flow available to both debt issuers and shareholders. It therefore starts from operating income (EBIT) instead of operating cash flow and it is taxed as if there is no debt.

$$\text{FCFF} = \underbrace{\text{Operating income} \times (1 - \text{Tax rate})}_{\text{Net operating profit after tax (NOPAT)}} + \text{Non-cash expenses} - \text{Reinvestment} - \quad (4)$$

Theorem 1.1 FCFF can be obtained from operating cash flow in the following way,

$$\text{FCFF} = \text{OCF} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Reinvestment}$$

Starting from the definition of OCF,

$$\text{OCF} = \text{NI} + \text{Non-cash expenses} - \Delta\text{OWC}$$

Net income is EBIT with tax and interest expensed away,

$$\begin{aligned} \text{NI} &= \text{EBIT} - \text{Interest expense} - \text{Tax expense} \\ &= (\text{Operating income} - \text{Interest expense}) \times (1 - \text{Effective tax rate}) \end{aligned}$$

and NOPAT is EBIT with only tax expensed away,

$$\text{NOPAT} = \text{Operating income} \times (1 - \text{Effective tax rate})$$

Thus, NI is NOPAT less a “taxed” interest expense.

$$\begin{aligned}
& \text{NI} = \text{NOPAT} - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\
\Rightarrow \text{OCF} &= \text{NOPAT} \\
& \quad - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\
& \quad + \text{Non-cash expenses} \\
& \quad - \Delta \text{OWC} \\
& = \text{FCFF} - \text{Interest expense} \times (1 - \text{Effective tax rate}) + \text{Reinvestment}
\end{aligned}$$

It should be noted that “non-cash expenses” often include items more than depreciation and amortization. All equations out there on the internet that uses only D&A as non-cash expenses cannot deal with real life cases. Take Parsons’ statement of cash flows in 2024 for example. As shown in Table 5, adjustments to reconcile net income to operating cash flow include deferred taxes and return on investments, and change in assets and liabilities include long-term liabilities which should be excluded from OWC.

Cash flows from operating activities	290,665
Adjustments to reconcile net income to net cash provided by operating activities	
Depreciation and amortization	99,251
Amortization of debt issue costs	7,799
Loss (gain) on disposal of property and equipment	948
Convertible debt repurchase loss	18,355
Deferred taxes	6,101
Foreign currency transaction gains and losses	6,919
Equity in losses (earnings) of unconsolidated joint ventures	23,361
Return on investments in unconsolidated joint ventures	40,162
Stock-based compensation	56,082
Contributions of treasury stock	59,778
Changes in assets and liabilities, net of acquisitions and newly consolidated joint ventures	
Accounts receivable	(163,139)
Contract assets	31,881
Prepaid expenses and other assets	35,830
Accounts payable	(42,686)
Accrued expenses and other current liabilities	79,984
Contract liabilities	(11,325)
Income taxes	(341)
Other long-term liabilities	(16,019)
Net cash provided by operating activities	523,606

Table 5: Statement of cash flows of Parsons in fiscal year 2024 in thousands.

Theorem 1.2 FCFF and FCFE can be associated in the following way,

$$\text{FCFF} = \text{FCFE} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Net borrowing}$$

Recall that FCFE is defined in terms of OCF, and we just demonstrated that FCFF can also be defined in terms of OCF,

$$\begin{aligned}
\text{FCFE} &= \text{Operating cash flow} - \text{Reinvestment} + \text{Net borrowing} \\
\text{FCFF} &= \text{Operating cash flow} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Reinvestment}
\end{aligned}$$

Therefore,

$$\begin{aligned}\text{Operating cash flow} &= \text{FCFE} + \text{Reinvestment} - \text{Net borrowing} \\ &= \text{FCFF} + \text{Reinvestment} - \text{Interest expense} \times (1 - \text{Effective tax rate}) \\ \Rightarrow \text{FCFF} &= \text{FCFE} + \text{Interest expense} \times (1 - \text{Effective tax rate}) - \text{Net borrowing}\end{aligned}$$

That is, FCFF is FCFE, plus the after-tax interests that goes to the debt issuers, minus cash from net debts the company raises.

Acquisition and working capital Companies that acquire subsidiaries frequently have their ΔOWC in CFS specifically written as “net of acquisitions” as shown in Table 5. As it turns out, the ΔOWC from balance sheet contains not only change in working capital due to internal operation, it also contains newly acquired working capital from acquired companies. This is because the ΔOWC in cash flow statements is used to arrive at operation cash flow, and cash used for acquisition is listed under the section of investing cash flow. As such,

$$\begin{aligned}\Delta\text{OWC (total)} &= \Delta\text{OWC (net of acquisition)} \\ &\quad + \text{The portion of acquisition cost that goes to working capitals}\end{aligned}$$

This means that our calculation of FCFF from NOPAT may be distorted, and FCFF based on NOPAT disagrees with FCFF based on OCF. For frequent acquirers, the definition of reinvestment must extend to include acquisition cost but also use ΔOWC from CFS to avoid double counting.

- $\Delta\text{OWC (total)} + \text{Reinvestment} + \text{Acquisition cost}$: double counting working capitals acquired from other companies
- $\Delta\text{OWC (net of acquisition)} + \text{Reinvestment}$: ignoring acquisition all together
- $\Delta\text{OWC (total)} + \text{Reinvestment}$: ignoring cost of fixed assets and goodwill in acquisition
- $\Delta\text{OWC (net of acquisition)} + \text{Reinvestment} + \text{Acquisition cost}$: agreeing with operating cash flow

Owner earnings Owner earnings is another variant of free cash flow popularized by Warren Buffett.

$$\text{Owner earnings} = \text{NI} + \text{Non-cash expenses} - \text{Maintenance capex} - \Delta\text{OWC}$$

The difference between FCFE and owner earnings is twofold:

1. Owner earnings does not include net borrowing; it disregards financing activities and focuses purely on operating activities
2. Owner earnings uses *maintenance capex* instead of capex

By using maintenance capex, the meaning of owner earnings becomes the free cash that is available to shareholders as well as available to invest in long-term assets. However, it is difficult to determine exactly how much capex is spent on maintaining existing assets. Some would use average of D&A for this figure, and the equation in effect becomes,

$$\text{Owner earnings} = \text{NI} + \text{Non-cash expenses less D\&A} - \Delta\text{OWC}$$

Invested capital and reinvestment rate Continuing on the idea of maintenance capex. If D&A is the proxy of maintenance cost necessary to keep the company running, whats left after we deduct maintenance cost from reinvestment expenses is the increase (or decrease) in total capital of the business.

$$\Delta\text{Invested capital} = \text{Capex} + \text{Purchase of intangibles} + \text{Acquisition cost} - \text{D\&A expenses}$$

Recall that the equation of FCFF can be expressed using NOPAT, non-cash expenses, and ΔOWC and reinvestment expenses, and recall that non-cash expenses include D&A expenses, as such,

$$\begin{aligned}\text{FCFF} &= \text{NOPAT} + \text{Non-cash expenses} - \Delta\text{OWC} - \text{Reinvestment} \\ &= \text{NOPAT} + \text{Non-cash expenses less D\&A} - \Delta\text{Invested capital}\end{aligned}$$

Therefore, NOPAT plus non-cash expenses less D&A represents all the cash inflow, and after we reinvest part of it to the operation, what is left is free cash that can be distributed to debt lenders or shareholders. Thus we can define *reinvestment rate* as such,

$$\text{Reinvestment rate} = \frac{\Delta\text{Invested capital}}{\text{NOPAT} + \text{Non-cash expenses less D\&A}}$$

Some literatures use a variant of definition where non-cash expenses less D&A is left out. This works if the only kind of non-cash expenses is D&A, which certainly does not apply to all companies.

2 Relative Valuation

Relative Valuation

Relative valuation, as opposed to absolute value models which directly calculate fair value, seeks to determine a company's worth by comparing its metrics and ratios with (1) other companies in the same sector or (2) the broad market as a benchmark.

Relative valuation refers to popular multiples or ratios that can be easily calculated from a company's financial numbers. This include,

- P/E ratio
- PEG ratio
- P/B ratio
- P/FCFE ratio
- EV/EBTIDA ratio
- EV/CFO ratio
- FCFF/EV ratio
- Debt-to-equity ratio
- Dividend yield
- Free cash flow yield
- Buyback yield
- Return on equity
- Return on invested capital
- Market implies growth appreciation period

Limitations of relative valuation Since relative valuation relies on multiples and ratios that only take a part of a company's financial statements into consideration, often they do not paint the whole picture and do not take any future events into consideration. By comparing multiples of ratios with companies or broad market, we also assume that the companies have the same marketing condition and cost structures. Lastly, relative valuation cannot be used on companies from different industries; different industries have different earning profiles.

Valuation multiples When multiples use the value measure of a company as its numerator, its called a *valuation multiple*. Depending on the type of value measure, multiples are categorized as either equity value

multiple or enterprise value multiple. The denominator of multiples are thus appropriate value drivers for each kind of value measure.

$$\text{Valuation multiple} = \frac{\text{Value measure}}{\text{Value driver}}$$

Value measure	Value driver	Valuation multiple
Equity value	Earnings	P/E ratio
	Book value	P/B ratio
	Free cash flow to equity	P/FCFE ratio
Enterprise value	EBITDA	EV/EBITDA ratio
	Revenue	EV/Revenue ratio
	Free cash flow to the firm	EV/FCFF ratio

Table 6: Valuation multiples using equity value and enterprise value as value measures and their respective appropriate value drivers.

2.1 Equity Value Multiples

P/E Ratio

Price-to-earnings ratio is the current market price of a company's share divided by its EPS. It is also called the price multiple or earnings multiple. P/E represents a the relative valuation of a stock and is best used when compared with past P/E ratios, with companies in the same industry, or with a broader market such as S&P 500 Index.

Since EPS is readily available in a company's income statement, P/E is the most popular choice of valuation. There are two kinds of P/E ratios depending on what kind of EPS value is being used:

1. Trailing P/E ratio: TTM EPS is used.
2. Forward P/E ratio: EPS value is based on estimated future earnings.

Forward P/E ratio uses estimated future earnings which vary greatly between analysts and are often overoptimistic. In terms of predicting the future performance, trailing P/E ratio is better than forward P/E ratio [10].

A P/E ratio below a certain value or above a certain value does not automatically make it undervalued or overvalued. A high P/E ratio could signal that the investors expect the company to grow, and a low P/E ratio could also mean that the company is facing difficulties. When EPS is close to zero or below zero, P/E ratio could fluctuate violently and become inapplicable.

P/E ratio does not mean the time it takes for investors to get back their revenue, despite it being a popular analogy. While a P/E ratio of 5 and share price of \$20 means that earnings are \$4 per year and it takes 5 years to earn \$20, it does not mean that the investor *gets* exactly \$20; it rather means that the company needs this much time to *earn* as much money as it is invested, and the money earned does not necessary go back to the investors.

Share buyback and P/E ratio P/E ratio can be manipulated by share buybacks since shares are bought with cash and such purchases are not considered expenses under generally accepted accounting practices. The result is reduced shares outstanding and shareholder's equity; in other words, higher P/E and higher ROE, assuming the exact same earnings and returns.

PEG Ratio

PEG ratio is obtained by dividing P/E ratio by estimated growth rate of EPS in percents,

$$\text{PEG} = \frac{\text{P/E ratio}}{\text{Estimated EPS growth rate}}$$

If the EPS of a company is posed to grow 20% and its current P/E ratio is 30, its PEG ratio would be $30 \div 20 = 1.5$.

As is the case of P/E ratio, the lower the PEG ratio is, the better it is. Based on the assumption that earnings growth is linearly proportional to current earnings, a PEG ratio of 1 is fair valued, and a PEG ratio below 1 is undervalued.

P/B ratio

Price-to-book ratio is obtained by dividing share price by *book value per share*. The book value of a company is derived from total assets minus total liabilities.

$$\text{Book value} = \text{Total assets} - \text{Total liabilities}$$

Book value is the amount of money the shareholders get when the company is liquidated. It is thus also known as *shareholder's equity*.

Book value is not a number found in financial statements, but its components are readily available in a balance sheet. In some cases, intangible assets and goodwill (the excess amount paid during a business acquisition over the fair value of acquired assets) are excluded from the calculation of P/B. In this case, the book value is known as *tangible* book value and the ratio becomes P/Tangible book value.

$$\text{P/Tangible book value} = \text{Total assets} - \text{Intangible assets and goodwill} - \text{Total liabilities}$$

A P/B ratio closer or below 1 is fair or undervalued as a rule of thumb. P/B ratio could be underselling companies that are heavy in intangible assets, such as service provider or software development companies, since the intangible assets they generate internally are not recognized on balance sheet (which only recognizes intangible assets that are acquired). Brands and patents all are intangible assets. Thus, P/B ratio is a reliable indicator only for the following industries:

- Capital intensive industries
 - Energy
 - Utility
 - Transportation
 - Large manufacturers
 - Mining and refining
 - Semiconductor manufacturing
- Financial services and banks

Banks are usually valued with P/B ratio instead of P/E ratio due to two main reasons. The first reason is also why P/B is favored for utility sector where companies often have a lot of assets as collaterals for loans.

1. Their earnings fluctuate often more than other industries due to federal interest and credit risks.

2. Banks often hold a large amount of loans and liabilities such as deposits and P/B ratio serve as a proxy of a bank's effectiveness on earning.

For example, Warren Buffett mentioned in one of shareholder meetings that he would buy back BRK.B shares when its P/B ratio is below 1.2. However, P/B ratio could still just be a proxy of intrinsic value of Berkshire Hathaway, as Warren did not buy back any shares in late 2024 when its P/B ratio is still under 1. At the beginning of April 2025 before a correction in the general equity market, P/B ratio of BRK.B reached a historical high of 1.1.

P/FCFE Ratio

Price-to-free cash flow to equity ratio is share price divided by FCFE per share.

$$P/FCFE = \frac{\text{Market capitalization}}{FCFE}$$

It measures how much at a premium the FCFE of a company is traded.

As is the case of P/E ratios, P/FCFE ratios are only meaningful when compared with historical P/FCFE values of a company, with the competitors of a company, or with the sector average.

2.2 Enterprise Value Multiples

Enterprise Value

Enterprise value (EV) is the total value of a firm, equal to how much it is needed to buy the whole company. It is calculated as such,

$$\begin{aligned} \text{Enterprise value} &= \text{Market capitalization} \\ &+ \text{Market value of interest-bearing liabilities} \\ &- \text{Cash and cash equivalents} \end{aligned}$$

where cash and cash equivalents include short-term investments such as T-bills and money market funds. Short-term investments are also otherwise known as *marketable securities*.

EV serves as a proxy of total capital that a company utilizes to generate profit. It is therefore suitable for other measures of profit at company level or in a wholistic approach, such as operating cash flow and EBITDA.

Negative enterprise value Asian companies tend to hoard a huge pile of cash, effectively rendering their EV negative. Cash at best only earns an interest no higher than bonds, so they are better spent elsewhere. An excess cash could therefore be a red flag that the management team is not good at capital allocation.

EV/EBITDA Ratio

EV/EBITDA ratio is EV divided by *earnings before interest tax depreciation and amortization* (EBITDA). EBITDA is considered a proxy of profitability or cash flow.

EBITDA is not available in income statement, but *earnings before interest and tax* (EBIT, also known as *operating income* OI) often is. EBITDA is simply EBIT plus non-cash expenses, i.e., amortization and

depreciation. If EBIT is not available, we can work from *net income* (NI) or revenue in the following manner:

$$\begin{array}{l}
 \text{NI} = \text{Revenue} \\
 \quad - \text{Cost of Goods Sold} \\
 \quad - \text{SG\&A} \\
 \quad - \text{R\&D} \\
 \quad - \text{Other OpEx} \\
 \quad - \text{Interest} \\
 \quad - \text{Taxes}
 \end{array}
 \left. \vphantom{\begin{array}{l} \text{NI} = \text{Revenue} \\ \text{Cost of Goods Sold} \\ \text{SG\&A} \\ \text{R\&D} \\ \text{Other OpEx} \\ \text{Interest} \\ \text{Taxes} \end{array}} \right\} \text{Operating Income (EBIT)} = \text{EBITDA} - \text{Depreciation} - \text{Amortization}$$

Use EV/EBITDA the same way as P/E ratio – compare present ratio with historical average, competitors, and sector.

Since EV/EBITDA does not take non-cash expenses and capital structure into consideration, it is suited for the following scenarios and sectors:

- Comparing companies within a same sector but with different debt levels
- Capital intensive sectors (e.g. energy) where debt level and depreciation is huge
- Sectors heavy on intangible assets (e.g. tech) where amortization is huge

A closely related ratio, EV/EBIT, is thus suitable for the following scenarios and sectors:

- Comparing companies with similar debt levels
- Financial firms with heavy debt obligation and tax impact, thus requiring EBIT which gives a clearer picture of capital structure than EBITDA

EV/FCFF Ratio

EV/FCFF is derived EV divided by *free cash flow to the firm* (FCFF).

2.3 Solvency Ratios

2.4 Yields

2.5 Return on Invested Capital

Return on Invested Capital

Return on invested capital (ROIC) is a measure of capital allocation and efficiency and defined as:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}}$$

There is no standard definition for "invested capital." Analysts often make a myriad of adjustments on invested capital to adjust for companies across various scenarios in various industries. The most general

formula of invested capital is:

$$\begin{aligned}\text{Invested capital} &= \text{Working capital} \\ &\quad + \text{PP\&E net of depreciation} \\ &\quad (+ \text{Intangibles net of amortization}) \\ &\quad (+ \text{Cash and its equivalents})\end{aligned}$$

Common adjustments of invested capital include:

- Inclusion/exclusion of intangibles; in particular, goodwill
- Capitalization of intangible investment
- Inclusion/exclusion of cash
- Time difference between NOPAT and invested capital

Working capital and excess cash Previously we have established that the “working capital” will always be OWC throughout our models – current assets plus interest-bearing debts minus current liabilities and cash.

A problem rises from the huge cash pile a company holds. Cash often is not part of core operation and just sits there earning a free interest rate, thus it serves more financing role than operating role. Companies keep a high level of cash due to several reasons:

- Young companies need a cushion for their operating activities
- Asian companies tend to hoard a huge pile as their part of culture

As a cashier keeps some cash in his register, *some* cash is necessary for operation. Analysts use 2% to 5% of revenue depending on the life stage of a company. We will use a conservative figure of 5%. Any cash more than that is stripped away in our calculation. Adding “operating” cash to OWC, we arrive at a variant of working capital close to what Aswath Damadoran uses [8]:

$$\begin{aligned}\text{OWC in the context of invested capital} &= \max(\text{Revenue} \times 0.05, \text{Cash and short-term investments}) \\ &\quad + \text{Accounts receivable} \\ &\quad + \text{Inventory} \\ &\quad + \text{Prepaid expenses} \\ &\quad - \text{Accounts payable} \\ &\quad - \text{Accrued expenses} \\ &\quad - \text{Other non-debt current liabilities}\end{aligned}$$

Intangible investment Prior to 2022, intangible assets are only recognized after merge and acquisition (M&A). Any internally generated intangibles are not recognized. Thus, companies such as Microsoft that are heavy on intangibles (brand, marketing, softwares, customer list, etc) appear to have less invested capital, which distorts ROIC.

When making adjustments to the calculation of ROIC, it is import to make sure that the numerator matches the denominator. In this particular adjustment, we are going to capitalize intangible investments

and correspondingly adjust amortization item in NOPAT. Start with adjusted NOPAT:

$$\begin{aligned}\text{Adjusted NOPAT} &= \text{NOPAT} \\ &+ \text{Reported amortization expense} \\ &+ \text{Intangible investment} \\ &- \text{Amortization of all intangibles}\end{aligned}$$

The reason why we add amortization back to NOPAT first is because part of intangible investment goes into maintaining and replenishing acquired intangibles, the same way capital expenditure covers purchases new PP&E and maintenance of existing PP&E. Leaving amortization in NOPAT and subtracting even more amortization from it runs the risk of double counting the amortization of acquired intangibles.

Intangible investment is not a GAAP item, and its subitems, research and development (R&D) and selling general and administration (SG&A), are often embedded in a big line item called operating expense. It is therefore difficult to estimate how much of operating expense goes into intangible investment, especially for companies that do not list R&D and SG&A. For those that do, we can roughly amortize them in the following way:

Item	% going to intangible investment	Useful live in years
R&D	100	6
Sales and marketing	70	2
General administration	20	2

Intangible investment is then added back to EBITA to reflect the fact that a portion of operating expense should be capitalized instead of expensed away. Finally we subtract newly calculated amortization from adjusted EBITA.

For invested capital, it will now have an item that is analog of net PP&E – net intangibles, accumulation of past intangible investments and amortization.

$$\begin{aligned}\text{Net intangibles} &= \text{Intangibles from last year} \\ &+ \text{Intangible investment} \\ &- \text{Accumulated amortization}\end{aligned}$$

Needless to say this adjustment is laborious. To get a good estimate of total intangibles of a company we need to go back at least 6 years. The matter is further complicated by laws such as Tax Cuts and Jobs Act (passed in 2017) and Tax Relief for American Families and Workers Act of 2024 which meddles in R&D capitalization.

Luckily the residual value of intangibles are (generally) assumed to be 0, so the total intangibles at year T is actually the weighted average of intangible investments in the past years.

$$\begin{aligned}\text{Net R\&D intangibles at year } T &= \text{R\&D}_T \\ &+ \text{R\&D}_{T-1} \times \frac{5}{6} \\ &+ \text{R\&D}_{T-2} \times \frac{4}{6} \\ &+ \dots \\ &+ \text{R\&D}_{T-5} \times \frac{1}{6}\end{aligned}$$

In an extreme case where R&D expenditure remains the same for the past 6 years, net R&D intangibles amounts to 3.5 times of R&D expenditure and R&D amortization is equal to R&D expenditure.

Goodwill Goodwill occurs in an M&A event when a company pays more than the fair value for an acquisition. Goodwill is added to the balance sheet and is never amortized away, but it is still subject to impairment tests every year. Say, a company has its fixed assets and intangibles valued at \$80 and is bought at \$100, the goodwill would be \$20. Write-downs will reduce goodwill item in balance sheet.

There are three ways to deal with goodwill:

1. Consider goodwill and goodwill impairment in vested capital
2. Include goodwill and *add back* goodwill impairment under the assumption that the value of goodwill does not wear out and management team must be held accountable for their acquisition decisions
3. Exclude goodwill from invested capital completely

For companies that did a one-time acquisition, it is best to leave goodwill out. For companies whose business model is to acquire other companies frequently, goodwill should be added to invested capital to see if management team overpaying for acquisition is justified. The latter companies usually have a large portion of their assets in goodwill and excluding goodwill will inflate their ROIC immensely.

Deferred tax assets and deferred tax liabilities DTAs and DTLs usually do not get into our calculation of invested capital. However, in cases where we use total assets for invested capital calculation, DTA gets in our way. Since DTA is just overpaid tax whose main purpose is to offset future tax payments, it produces nothing and serves no operating purpose, thus it should be excluded.

Invested capital calculation We now return to calculate invested capital. We can make a myriad of adjustments here and there, but the most important part is that we are consistent with our numerators and denominators and we are consistent in the way we treat intangibles and working capital across companies across industries. Our calculation is outlined below:

- Intangible investment is adjusted when R&D is recorded as an expense under operating expenses and R&D intangibles are not reported in the section of intangible assets
- Goodwill is included and unimpaired
- “Net working capital” is defined as OWC plus operation cash (no more than 5% of revenue)
- DTA is removed

This leads to the formula of invested capital:

$$\begin{array}{lcl}
 \text{Invested capital} = & \left. \begin{array}{l} \text{Current assets} \\ + \text{Net PP\&E} \\ + \text{Goodwill} \\ + \text{Intangibles} \\ + \text{Other long-term assets} \end{array} \right\} & \text{Total assets} - \text{DTA} \\
 & - \text{Current liabilities (less short-term debts and principle parts of long-term debts)} & \\
 & - \text{Excess cash} &
 \end{array}$$

which means that we only need total assets, current liabilities less debts, (DTA if we are feeling fancy), and revenue to calculate invested capital.

Time difference Assets require time to generate revenue. Newly purchased assets do not return anything right away. The most common time difference used between NOPAT and invested capital is 1 year, meaning that, to calculate ROIC at the end of year 2021, we will use NOPAT at the end of 2021 and assets at the end of 2020. Some sources use *average* invested capital, which is the average of invested capital at the beginning of the period and invested capital at the end of the period.

Example We will use NVIDIA as an example. Financial statements of NVIDIA are listed in Table 7 and Table 8. First we will calculate NOAPT. Using income tax of \$11,146 million and pretax income of \$84,026 million, we get effective tax rate of 13.26%. Using operating income of \$81,453 million, NOPAT is thus \$70,648.

NVIDIA only include acquired intangibles and patents in its intangible item in balance sheet, so it is eligible for intangible investment adjustment. Using R&D and SG&A expenditures from Table 8, we have \$25,477 million of net intangibles from R&D and SG&A ending 2024. The amortization expense in 2025 is 1/6 of R&D expense from 2019 to 2024 and 1/2 of SG&A in 2024, which amount to \$6,395 million. Additionally, amortization expense associated with acquired intangibles and patents is \$593 million as listed on page 65 of 10-K form issued in 2025.

Now look at the balance sheet. NVIDIA's TWC in 2024 is \$12,582 million and its NWC is \$33,714 million. Net PP&E and other assets, not including intangibles, sums up to \$9,760 million. Revenue in 2024 is \$60,922 million, which means that \$4,233 million of cash exceeds 5% of revenue. The exact number of invested capital depends on which adjustment we want to make – to capitalize R&D and SG&A, to use TWC or NWC, to exclude excess cash, or to exclude goodwill.

Revenue	130,497	Current assets:	
Gross profit	97,858	Cash and cash equivalents	7,280
Operating expense		Marketable securities	18,704
Research and development	12,914	Accounts receivable, net	9,999
Sell, general and administration	3,491	Inventories	5,282
Total operating expense	16,405	Prepaid expenses and other current assets	3,080
Operating income	81,453	Total current assets	44,345
Interest income	1,786	Properties and equipment, net	3,914
Interest expense	(247)	Operating lease assets	1,346
Other, net	1,034	Goodwill	4,430
Other income (expense), net	2,753	Intangible assets, net	1,112
Income before income tax	84,026	Deferred income tax assets	6,801
Income tax expense (benefit)	11,146	Other assets	4,500
Net income	72,880	Total assets	65,728

(a) Excerpt of NVIDIA's income statement of FY2025, in millions.	Current liabilities:	
	Accounts payable	2,699
	Accrued and other current liabilities	6,682
	Short-term debt	1,250
	Total current liabilities	10,631
	Total liabilities	22,750

(b) Excerpt of NVIDIA's balance sheet of FY2024, in millions.

Table 7: Financial statements of NVIDIA.

Table 9 shows ROIC based on various adjustments discussed above. For reference, gurufocus.com gives 174.69% while financecharts.com gives 132.62%, both of which falls in our range of 316% to 108%.

ROIIC When a company is in its maturing stage with a large amount of NOPAT and invested capital accumulated, return on newly invested capital is muted. A company may had a good capital allocation in the majority of its history and its recent blunders can be thus muted, as shown in Figure 3. *Return on incremental invested capital* is thus a measure of recent capital allocation.

FY	R&D	SG&A
2025	12,914	3,491
2024	8,675	2,654
2023	7,339	2,440
2022	5,268	2,166
2021	3,924	1,940
2020	2,829	1,093
2019	2,376	991
2018	1,791	815

Table 8: R&D and SG&A expenditures of NVIDIA since fiscal year 2018 in millions.

Without in-house intangibles	TWC	NWC, excluding excess cash	NWC, including excess cash
Excluding goodwill	316.23%	180.05%	162.52%
Including goodwill	263.90%	161.78%	147.49%
With in-house intangibles	TWC	NWC, excluding excess cash	NWC, including excess cash
Excluding goodwill	166.98%	123.38%	115.81%
Including goodwill	152.82%	115.48%	108.81%

Table 9: ROIC of NVIDIA in fiscal year 2025.

Return on Incremental Invested Capital

Return on incremental invested capital (ROIIC), or return on new invested capital (RONIC), is the ratio of change in NOPAT to change in invested capital over a period of time.

$$\text{ROIIC}(t, k) = \frac{\text{NOPAT}_t - \text{NOPAT}_{t-k}}{\text{IC}_{t-1} - \text{IC}_{t-k-1}}$$

Time period k is often 3 to 5 years to smooth out year-to-year fluctuations.

Since ROIC reverts to 10% in perpetuity and ROIC approaches ROIIC asymptotically, ROIIC can differ from ROIC during growth stage and approximates 10% in stable stage.

ROIC and growth ROIC itself measures capital efficiency, but it is not necessary the rate at which the company grows. As is the case of Asian companies, even if they have very efficient capital allocation and utilization, they won't grow if they don't reinvest their huge pile of cash. As such, the growth of NOPAT is co-determined by ROIC and reinvestment rate. We use ROIIC when the company is in growth stage so as to not estimate NOPAT growth using the whole history of capital allocation and underweight recent management decisions.

$$\text{Estimated NOPAT growth} = \frac{\text{NOPAT}_{t+1} - \text{NOPAT}_t}{\text{NOPAT}_t} = \text{ROIIC}(t, k) \times \text{Reinvestment rate}_t$$

The relationship becomes clear if we expand the definitions of ROIIC and reinvestment rate. Essentially, we are predicting NOPAT growth by applying past ROIIC to IC increase at the current year. This, of course, assumes that ROIIC remains constant in the coming years.

$$\text{ROIIC}(t, k) \times \text{Reinvestment rate}_t = \frac{\text{NOPAT}_t - \text{NOPAT}_{t-k}}{\text{IC}_{t-1} - \text{IC}_{t-k-1}} \times \frac{\text{IC}_t - \text{IC}_{t-1}}{\text{NOPAT}_t}$$

Take, again, NVIDIA for example. Given its capex, investments, depreciation and amortization, and

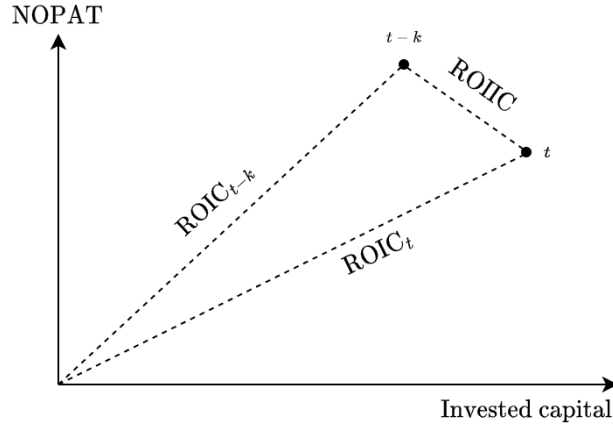


Figure 3: ROIC and ROIC at two different times t and $t - k$, showing that ROIC remains positive in both years while ROIC is negative.

change in NWC listed in Table 10, we conclude that its reinvestment rate is 50.48%. Combined with the ROIC of 115.48%, its NOPAT is expected to grow at 58.3%.

Capex	3,236
Intangible investment	16,405
Depreciation	1,300
Amortization of in-house intangibles	6,395
Δ NWC	28,365
NOPAT adjusted for in-house intangibles	79,849
Reinvestment rate	50.48%
ROIC,	
adjusted for in-house intangibles,	
using NWC less excess cash,	115.48%
including good will	
Expected NOPAT growth	58.3%

Table 10: Reinvestment rate and expected NOPAT growth of NVIDIA at the end of fiscal year 2025. Numbers are in millions except for percentages.

When depreciation and decrease in NWC is greater than capital expenditure, reinvestment rate goes negative. Two main reasons why a company has negative reinvestment are,

1. The company over-invested in the past and is living off past investment.
2. The company has huge increase in short-term debts due to long-term debts maturing soon.
3. The company is liquidating itself and letting assets wear down.

For the first case, we should use industry average. For the third case, however, there is no growth to project. The company is digging its grave.

For the second case, we can take Parsons for example, whose debts are listed in Table 11. Some debts are due in 2025 so they are reclassified from long-term debts to short-term debts. At the same time more convertible notes are issued, so the net borrowing is positive. An increase in short-term liabilities (and hence current liabilities) means a decrease in cash flow, but remember that,

- FCFF is the cash flow available for both debt issuers and shareholders

- FCFE has a term of net borrowing

Therefore, when we calculate the working capital for FCFF or FCFE or invested capital, including this increase in short-term debts is double-counting and distorts the figures. Table 5 also did not consider this increase in short-term debts when it reconciled net income to operating cash flow – the item “accrued expenses and other current liabilities” shows a increase of \$79,984 which is far lower than increase in short-term debts while accrued expenses also increases, and issuance and repurchases of convertible notes are under “cash flows from financing activities” section.

	2024	2023
Short-Term Debt:		
Delayed draw term loan	350,000	–
Convertible senior notes due 2025	113,405	–
Total Short-Term Debt	463,405	–
Long-Term Debt:		
Delayed draw term loan	–	350,000
Convertible senior notes due 2025	–	400,000
Convertible senior notes due 2029	800,000	–
Debt issuance costs	(15,904)	(4,037)
Total Long-Term Debt	784,096	745,963
Total Debt	1,247,501	745,963

Table 11: Debt and credit facilities (footnote 11) of Parsons in fiscal year 2024 and 2023 in thousands.

2.6 Return on Equity

Return on Equity

Return on equity (ROE) measures how much earnings a company generates from shareholders’ equity.

$$\text{ROE} = \frac{\text{Net income}}{\text{Shareholders' equity}}$$

Equity Reinvestment Rate

Equity reinvestment rate (ERR) measures how much of earnings is reinvested back to capital.

$$\text{ERR} = \frac{\text{Capex} + \Delta\text{OWC} - \text{Depreciation} - \text{Net debt issued (repaid)}}{\text{Net income}}$$

ROE and ERR are analogues of ROIC and RR on the equity side. Likewise, cost of equity (CoE) is used in the place of WACC. Different from NOPAT and ROIC which ignores capital structure, on the equity side, we must take capital structure into consideration as well as many other equity side shenanigans:

1. Earnings can be raised by issuing shares.
2. Earnings can be raised by issuing debt.
3. Leverage (total assets to shareholders’ equity) inflates ROE.

While the effect of share dilution can be eliminated by calculating EPS, the effect of debt financing must be accounted by adjusting net debt issued (repaid) in equity reinvestment rate.

ROE and P/B Since ROE and P/B are both based on shareholder's equity, they are related in the following fashion,

$$\text{ROE} = \frac{\text{P/B}}{\text{P/E}}$$

2.7 Cash Return Metrics

The return metrics we have been talking about can be made on a cash basis. For cash return metrics we will focus on OCF instead of earning figures (NOPAT or NI). Non-cash expenses such as D&A are stripped away. SBC is still excluded from our definition of OCF. Thus *cash return on invested capital* and *cash return on equity* can be defined.

Cash Return on Invested Capital

Cash return on invested capital (CROIC) is the cash equivalent of ROIC where we use FCFF in place of NOPAT.

$$\text{CROIC} = \frac{\text{FCFF}}{\text{IC}}$$

Cash Return on Equity

Cash return on equity (CROE) is the cash equivalent of ROE where we use FCFE in place of NI.

$$\text{CROE} = \frac{\text{FCFE}}{\text{Shareholders' equity}}$$

Cash return metrics can be derived by multiplying their non-cash counterparts with $1 - \text{reinvestment rate}$ since free cash flow is what is left after the reinvestments are made.

$$\text{CROIC} = \frac{\text{FCFF}}{\text{IC}} = \frac{\text{NOPAT}}{\text{IC}} - \frac{\text{Net reinvestment}}{\text{IC}} = \frac{\text{NOPAT}}{\text{IC}} \cdot (1 - \text{Reinvestment rate})$$

As for value creation, it can be argued that, instead of using ROIC and ROE, their cash equivalents should be compared to WACC and CoE respectively. This is unnecessary strict and will be shown in the next section where we introduce discounted cash flow model to connect ROIC and WACC.

2.8 Determinants of Ratios

3 Fair Value Calculation

Based on the EMH that the price accurately reflects all the cash flow in the future, we can project cash flow of a company, discount future cash flow by some discount rate, and arrive at a "fair" price for its shares. The calculation of fair value is thus similar to that of bonds; in the case of valuing a company, we replace par with *terminal value* and coupon with *free cash flow* in the equation. This method of valuing a company is thus called *discounted free cash flow model* (DCF).

Other valuation models exist such as:

- Reverse discounted free cash flow model

- Dividend discount model
- Excess returns model

Since DFC is not suitable for all companies, these models are necessary valuation tools. For example, dividend discount model are suitable for companies that pay out a great portion of their cash flow as dividends; excess returns model suit financial companies such as banks and insurance [17].

3.1 Discounted Cash Flow Model

The calculation of DFC is based on the following four components:

- Time horizon N
- Free cash flow projection, with the following varieties of cash flow available:
 - Free cash flow *to the firm*, FCFF, or *unlevered* free cash flow
 - Free cash flow *to equity*, FCFE, or *levered* free cash flow
- Discount rate r , with the following choices available:
 - Weighted average cost of capital (WACC), which is appropriate for FCFF
 - Cost of equity (CoE) or required rate of return (RRR), which is appropriate for FCFE
 - Custom discount rate such as 10-year treasury bond yield, GDP growth rate, or CAGR of S&P 500, sometimes with a margin of safety applied
- Terminal value (TV), with the following models available:
 - Gordon growth model, which uses a perpetuity growth rate g
 - Exit multiple approach, which assumes a valuation multiple at the end of time horizon

The present value (PV) of future cash flow FCF_i at year i is discounted by the power of $1 + r$.

$$PV_i = \frac{FCF_i}{(1 + r)^i}$$

The sum of PVs of future FCFF and PV of TV is *enterprise value* (EnV). The sum of PVs of future FCFE and PV of TV is *equity value* (EqV). EnV and EqV are related in the following way:

$$EnV = \sum_{t=1}^N \frac{FCFF_t}{(1 + r)^t} + \frac{TV}{(1 + r)^N} \quad (5)$$

$$EqV = \sum_{t=1}^N \frac{FCFE_t}{(1 + r')^t} + \frac{TV}{(1 + r')^N} \quad (6)$$

$$= EnV - \text{Total debts} + \text{Cash and its equivalents and marketable securities} \quad (7)$$

Free cash flow projection Free cash flow projection contains a lot of bias and guesswork that a consensus between various analysts must be used; if there is no estimation of future free cash flow from analysts available, an approach is to interpolate free cash flow growth rate starting from the mean of last reported growth rate and historical average growth rate and to reduce it gradually and at the end of time horizon it reaches stable growth rate g , which is often assumed to be the yield of 10-year government bond, or an average of the yield of 10-year government bond for the last 5 years [17].

Terminal value In Gordon growth model, the same stable growth rate g is also used to calculate TV. Given the stable growth rate g , the appropriate discounting rate r for FCF, the model gives an approximate value of TV,

$$TV = \frac{FCF_N \cdot (1 + g)}{r - g} \quad (8)$$

TV can be otherwise estimated by multiplying the earnings at the end of time horizon with an exit multiple. For FCFF, TV would be the product of estimated EBITDA and estimated EV/EBITDA. For FCFE, TV would be the product of estimated net income and estimated P/E ratio. Exit multiples are often derived from multiples of comparable companies and multiples of precedent transactions of comparable targets.

Equity value The last step of DCF is to subtract net debt (debt D minus cash and cash equivalents C minus marketable securities M) from enterprise value. The results is called *equity value* (EqV) which represents the calculated fair value of the company's equities. Divide this number by shares outstanding and we arrive at the "fair" price of a share.

$$\text{Fair share price} = \frac{\text{EqV}}{\text{Shares outstanding}} = \frac{\text{EnV} + M + C - D}{\text{Shares outstanding}}$$

3.1.1 Required Rate of Return

Required rate of return (RRR) is an import part of DCF models since future cash flows are discounted by this rate. The further out in the future a cash flow is, the more it is discounted. Terminal value in particular is heavily discounted as it is the furthest out in the future.

Instead of discounting cash flow, we can also see it the other way. Given a required rate of return, I expect each dollar I receive to grow exactly at the same rate and compound as time goes on. Therefore a dollar at present should worth more in the future, and a dollar in the future is worth less at present.

Depending on which kind of cash flow we are using, there are two RRRs:

- FCFF is discounted by *weighted average of cost of capital* (WACC)
- FCFE is discounted by *cost of equity* (CoE)

Cost of equity Cost of equity is what shareholders (you and me) requires a company to return for each dollar they invest. The required return consists of two parts:

1. Risk premium in excess of risk-free rate that the shareholders require to be compensated for taking risk. In plain English, the risky it is, the more I should get paid for investing in it.
2. Opportunity cost of each dollar invested into the company instead of getting market return. If I can get 10% return out of S&P 500, why would I invest in a company that underperforms?

To put these together, the cost of equity can be expressed as the formula below,

$$\text{CoE} = \min(r_f + \beta r_p, 10\%)$$

where r_f is risk-free rate (10-year US treasury bond, or any long-date bond in the same country as the company) and r_p is risk premium (about 6% for US stocks).

The beta coefficient is a measure of sensitivity. There are a myriad ways to get beta, two of which are,

1. Measure how sensitive the stock price is related to the broad market (market beta)
2. Calculate it from fundamental financial figures (e.g. ROA, ROE, earnings) of the company (accounting beta)

Popular methods of beta calculation involve some sort of regression. It never has financial figures incorporated directly into the coefficient itself, so the coefficient, obtained this way, is actually a measure of *sensitivity* instead of *risk*.

It is a wide spread idea that the risks of assets are captured and reflected in their pricing, *i.e.*, volatile price = risky. Since the market-portfolio (such as VOO) only have systematic risks that cannot be diversified away, if we accept the assumption that volatility = risk, the sensitivity of a stock would capture both systematic risk and idiosyncratic risk, and a beta of greater of 1 would thus be considered “risky”.

However, this “volatility = risk” notion is very wrong because of the very nature of beta itself as a statistic measurement instead of qualitative factor. Beta as a regression coefficient clearly is based on two factors:

1. Variance of the asset price
2. Correlation between the asset and the market

Thus, it is possible to have a low beta stock whose price fluctuates like a rollercoaster and somehow maintains a low correlation with the market. It can also have 0% or even negative overall return and still have a deceptively low beta because beta itself holds no implication of return whatsoever.

Accounting beta is derived by regression the firm’s return metrics or earnings against the average of other comparable firms. While it seems more accurate to the real picture, it has smaller sample size and is more prone to error, not to mention the enormous labor required to collect all the data. Still, Aswath Damodaran maintains a website that lists unlevered and levered beta calculated in this fashion [6].

If we treat “beta” as a number that represents the “risk” that comes from leverage a company takes to amplify its returns, we can at least identify two kinds of leverages from the financial figures of a firm:

1. Operating leverage: the higher proportion fixed cost is compared to variable cost, the more operating profit a company generates when its revenue goes up, but at the same time the more loss it suffers when revenue goes down.
2. Financial leverage: the more debt a company takes, the more money it can invest in capitals and even pay dividends, and at the same time failing to repay debt in a timely manner would bankrupt the company.

As such, the “leverage” beta would be calculated as the product of operating leverage and financial leverage. Operating leverage is approximated by the ratio of % change in operating income to % change in revenue, and financial leverage is $1 + \text{ratio of market value of debt to equity with tax deducted}$.

$$\beta = \frac{\% \text{ change in operating income}}{\% \text{ change in revenue}} \cdot \left(1 + \frac{D}{E} \cdot (1 - \text{Tax rate}) \right)$$

Using the average of the market¹ and the risk-free rate of 4%², we can work back from historical 10% return of the market and get the average leverage beta and risk premium corresponding to this average leverage beta.

$$10\% = 4\% + 0.97 \cdot 1.13 \cdot \text{Risk-free premium} \Rightarrow \text{Risk-free premium} = 5.5\%$$

If a firm has leverage beta of 2, we would require it to return $4\% + 2 \times 5.5\% = 15\%$. Indeed it does not capture other leverages a firm can take, such as political leverage. It does also not capture other non-leverage risks such as change in macroeconomics, general market sentiment, and price of raw materials.

Cost of debt Just like shareholders require higher returns from risky firms, lenders also require higher yield from debts of risky firms. There are two ways to calculate how much lenders are demanding their coupons:

1. $\text{CoD} = \text{Interest expense} \div \text{Book value of debt}$

¹Operating leverage = 0.97 and D/E = 1.13. Source.

²10-year Treasury note is approximately 4.38% as of June 6, 2025.

2. CoD = YTM of corporate bonds
3. CoD = Risk-free rate + Implied credit spread

The first approach is the easiest way as interest expenses and book value of debts are readily available in financial statements, but it also gives an inaccurate figure of how much lenders are demanding if the company wants to raise debts *today*. The figure produced by the first approach is the *coupon rate* – the nominal yield paid in fixed amount. The second approach is *yield to maturity* (YTM), the yield inferred from the market price of bonds.

For fixed rate bonds that circulate in secondary markets, bonds are priced according to treasury yield (cost of opportunity) and default risk. If a company has more default risk, its bonds have less demand, pushing price lower, and heighten its yields inferred from price, thus higher YTM. If the company wants to raise more debt, it would have to raise its coupon rate to attract demand. For example, Procter & Gamble issued \$516m bonds at a fixed coupon rate of 3.55% which mature at 2040. These bonds are traded at 82.5% of its face value of \$1,000 as of Jun 6th 2025, thus having YTM of 5.18%.

For companies that issue a lot of debts each of which has vastly different YTM and term to maturity (TTM), it can still be difficult to calculate overall cost of debt. In this case, we will use an approximated YTM based on average market value of bonds and average term to maturity using the following formula,

$$\text{YTM} = \frac{C + (F - P)/n}{(F + P)/2}$$

where C is coupon payment (= interest expenses), F is book value of bonds (sum of face value of debts), P is market value of bonds, and n is average of term to maturity weighted by book value. Table 12 lists 5 most outstanding debts of Verizon since it has 334 symbols in the bond market. Using the 100 most outstanding debts, $C = \$4.46$ billion (an average coupon rate of 3.7%), $F = \$120.59$ billion, $P = \$88.29$ billion, $n = 12.41$ years, and YTM amounts to 6.76%. If we calculate using just interest expense of \$6.649 billion and book value of debts of \$171.695 billion, we would get 3.87% instead, which is (close to average coupon rate and) way below what the market demands.

Symbol	YTM	Price (% of par)	Coupon (%)	TTM (yr)	Outstanding amt (B USD)
VZ5252609	5.42	84.80	2.36	6	4.51
VZ5252609	4.77	97.19	4.02	4	3.96
VZ5181143	7.33	58.71	2.99	31	3.65
VZ4719974	4.55	99.35	4.33	3	3.62
VZ5148414	6.76	67.69	3.70	35	3.04

Table 12: Five most outstanding debts of Verizon with non-zero volume in the secondary market as of June 8th, 2025.

For companies whose bonds are in the banks and not publicly traded, we cannot get YTM from its bonds. Another way is to infer it from interest coverage ratio:

$$\text{Interest coverage ratio} = \frac{\text{Operating income}}{\text{Interest expense} - \text{Interest income}}$$

The less a company can cover its (net) interest expenses, the more default risk it presents, and thus lenders demand more yield. We can then infer credit spread from Aswath's data [?]. Take Verizon for example again, with its operating income of \$30.9 billion and interest expense of \$6.65 billion found in I/S, its interest coverage ratio of 4.64 would land us at %0.85 spread. Even with 10-year treasury note at 4.5%, this would give us 5.35%, lower than the actual YTM from its bonds traded in the market, but still way closer than 3.87% calculated from book value.

Market value of debt WACC uses *market value* of debts and equity, NOT book value. Previously we have seen that if a company has bonds in the secondary market, the market price of those bonds are also available and its total market value of bonds is simply the sum of bond price times outstanding amounts. If a company has too many debts, as is the case of Verizon, market value of debts can be estimated using CoD, weighted TTM, and book value of debts.

$$\text{Market value of debt} = \text{Interest expense} \times \frac{1 - \frac{1}{(1+\text{CoD})^n}}{\text{CoD}} + \frac{\text{Book value of debts}}{(1 + \text{CoD})^n}$$

Thus the market value of debts of Verizon is \$130.84 billion. With the market capitalization of Verizon being \$168.343 billion at the end of 2024, its D/E ratio based on market value is 0.777, way lower than 1.707 if we calculate the ratio based on book value as shown in Table 13.

	Debt	Equity	D/E ratio
Book value	171.695	100.575	1.707
Market value	130.84	168.343	0.777

Table 13: Book value and market value of debt and equity on Dec 31st, 2024.

Weighted average cost of capital WACC measures the expected payment on average to all its security holders (stockholders and debt issuers) to finance its assets. Since companies are financed on multiple sources such as stocks, debts, subsidies, WACC measures the minimum rate of return to satisfies their investors. Suppose that a company is financed solely by debts and stocks,

$$\text{WACC} = \frac{E}{D+E} \times \text{CoE} + \frac{D}{D+E} \times \text{CoD} \times (1 - \text{Tax rate}) \quad (9)$$

Take Verizon again for example. Its leverages, beta, CoE, and CoD are all listed in Table ???. Its WACC turns out to be 14.69%, higher than 10% of total market since, mostly contributed by its high operating leverage.

% change in revenue	0.608
% change in operating income	2.088
Operating leverage	3.43
Market value D/E	0.777
Effective tax rate (%)	21.9
Financial leverage	1.17
Leverage beta	4.01
Cost of equity	26.07%
Cost of debt	6.76%
WACC	14.69%

Table 14: Calculation of WACC of Verizon using financial figures at Dec 31st, 2024.

3.1.2 Example

Example Take NVIDIA for example. We will be using data from 2020 and calculate its cash flow projection as well as fair value based on our assumptions as list in Table 14. For beta we will use the value from 2021 since beta from previous periods are unavailable as of 2025.

We start from levered beta β :

$$\beta = 0.96 \times (1 + 0.7 \times 11898 \div 12204) = 1.615$$

Item	Value
Historical average growth rate of free cash flow to equity	60%
Free cash flow (2020/01/31)	\$4,272
Free cash flow annual growth rate (2020/01/31)	36%
Interest expense (2020/01/31)	\$52
Total liability D (2020/01/31)	\$11,898
Shareholder equity E (2020/01/31)	\$12,204
Cash, cash equivalents, and marketable securities C	\$10,897
Shares outstanding (2020/01/31)	618
Unlevered beta for semiconductor industry β_0	0.96
Equity risk premium (2020)	4.72%
5-year average of 10-year U.S. treasury (2016 Jan. – 2020 Jan.) $r_f = g$	2%
Time horizon N	10 years

Table 15: Data from NVIDIA 2020 financial statement as well as other sources [2] [7] [11]. All numbers are in millions except for percentages and time horizon.

Cost of equity R_E is thus:

$$R_E = 0.02 + 1.615 \times 0.0472 = 0.0962$$

To calculate projected free cash flow, we must first obtain FCFE growth rate g_i . The growth rate of FCFE at first year g_1 will be the mean of historical average 60% and last growth rate 36%, which amounts to 48%. The growth rates on the last year of time horizon g_{10} equals to risk free rate of 2%. Growth rates for the years in the middle will be interpolated using these two numbers. Growth rates and FCFE and discounted FCFE for each year is listed in Table 15. The sum of all discounted FCFE is thus \$112,676 million.

Year i	Growth rate g_i (%)	FCFE $_i$ (million \$)	Present value of FCFE $_i$ (million \$)
1	48	6,322	5,767
2	38.8	8,775	7,303
3	34.2	11,777	8,940
4	29.6	15,263	10,570
5	25	19,078	12,053
6	20.4	22,970	13,238
7	15.8	26,600	13,984
8	11.2	29,579	14,186
9	6.6	31,531	13,795
10	2	32,162	12,836

Table 16: FCFE and FCFE growth rate according to DFC assumptions.

Terminal value will be based on the last FCFE which is \$12,836 million.

$$TV = \frac{12,836 \times 1.02}{0.0962 - 0.02} = 171,820$$

$$\text{Present value of TV} = 171,820 \div 1.0962^{10} = 68,576$$

Finally, we arrive at enterprise value and equity value. Since our calculated enterprise value is way larger than cash and debt at 2020, we ignore them and directly derive “fair” share price from enterprise value.

$$EqV \approx EnV = 112,676 + 68,576 = 181,252$$

$$\text{Fair share price} = 181,252 \div 618 = 293.29$$

On Jan. 20th 2020, NVIDIA trades at \$250.4 per share (this price level was before a 4:1 split in 2021 and a 10:1 split in 2024). Calculation shows that this price level is below fair value; that is, NVIDIA is undervalued.

Limitations of DCF It goes without saying that DCF as a prediction model is as good as its assumptions and its inputs. The projection of cash flow could be wrong, the CAPM we used in the calculation of discount rate is also empirically wrong, and the market can simply disregard a calculated “fair” value. Common practices include

- Slap a 20% margin of error to the calculated fair value; that is, if the current share price is lower than 20% of calculated fair price, we can confidently say that the company is undervalued.
- Prepare different FCFE projections; the best case, the worst case, and the normal case..

3.2 Reverse Discounted Cash Flow Model

Since FCF projection remains the most uncertain and the most influential part of FCF, we can instead start from the present-day share price and reverse engineer the required cash flow to sustain that price level. If the result required future cash flow is unrealistically high, it implies that the current share price is too high and it prices more about hype than future cash flow.

Modeling Each Year’s FCFE Reverse-engineering future cash flow implies that there is only one variable among future cash flow. The most common assumption to make is that the FCFE growth remains constant over the years ($g_1 = g_2 = \dots = g_{N-1}$). We can also assume that the growth rate starts from an unknown value g_1 and declines at the same rate over the years until it reaches terminal growth rate g , the same assumption we did previously for DCF. Thus,

$$\begin{aligned} \text{For } i = 2 \dots N, g_i &= g_1 \cdot \frac{N-i+1}{N} + g \cdot \frac{i-1}{N} \\ \text{For } i = 1 \dots N, \text{FCFE}_i &= \text{FCFE}_0 \prod_{j=1}^i (1+g_j) \\ \text{TV} &= \frac{\text{FCFE}_N \cdot (1+g)}{R_E - g} = \frac{1+g}{R_E - g} \cdot \text{FCFE}_0 \prod_{j=1}^N (1+g_j) \end{aligned}$$

where FCFE_0 is the last reported FCFE.

With all variables depending solely on g_1 , we can use trial error to approximate g_1 that matches a share price of \$250.4, a terminal growth rate of 2%, and a CoE of 9.62%, the same set of data and assumptions based on Table 14. The value of g_1 turns out to be 32.9%, which is below last reported FCFE growth rate. If one expects FCFE of NVIDIA to grow at the same pace as last reported, the FCFE growth of 32.9% implied by current share price would be way below his expectation, and NVIDIA could be undervalued.

3.3 ROIC/ROE and Discounted Cash Flow

Recall that, when we calculate terminal value, Gordon growth model calls for a growth rate. Previously we also learned that growth is tied to ROIC/ROE and reinvestment rate (RR). It is therefore crucial to incorporate ROIC/ROE into our DCF model as it gives us more insight. Let us preface our analysis of ROIC/ROE incorporation with the following assumptions and generalizations:

- Our DCF model consists of only terminal stage for now. The company is in a mature stage and grows at a perpetual growth rate of $g = \text{ROIC} \times \text{RR}$.
- Interest ratio I and effective tax rate T remain the same so that EBIT, NOPAT, and earnings E grow at the same rate. In particular, $E = \text{NOPAT} \times (1 - I) = \text{EBIT} \times (1 - I) \times (1 - T)$.

These assumptions allow us to apply growth (and eventually ROE) to earnings and convert it to free cash flow conveniently. Let E' be the earnings at next year and $FCFE'$ be the FCFE at next year. Since FCFE is what is left after reinvestments are made, we have,

$$\begin{aligned} FCFE' &= E' \times (1 - RR) \\ &= E' \times \left(1 - \frac{g}{ROE}\right) \\ &= E \times (1 + g) \times \left(1 - \frac{g}{ROE}\right) \end{aligned}$$

Gordon growth model determines the value of a company based on the free cash flow next year. Therefore, given the required rate of return r ,

$$\text{Value} = \frac{FCFE'}{r - g} = \frac{E \times (1 + g) \left(1 - \frac{g}{ROE}\right)}{r - g}$$

Dividing the value by earnings E gives a “fair” P/E ratio,

$$\text{Fair P/E ratio} = \frac{\text{Value}}{E} = \frac{(1 + g) \left(1 - \frac{g}{ROE}\right)}{r - g}$$

Based on different values of growth g and ROE spread (ROE minus required rate of return), we get a fair P/E ratio matrix as shown in Table 16. Notice that when ROE spread is 0%, fair P/E ratio barely moves as growth varies. When ROE spread is negative, the more growth the company pursues, the less P/E ratio it becomes even to the point where it could go negative. This is because, with an inefficient capital utilization, the company must reinvest more than it can afford to achieve a high growth.

Growth \ ROE spread	ROE spread	-3%	-2%	-1%	0%	1%	2%	3%
	2%	9.11	9.56	9.92	10.20	10.43	10.63	10.79
	3%	8.41	9.20	9.81	10.30	10.70	11.04	11.32
	4%	7.43	8.67	9.63	10.40	11.03	11.56	12.00
	5%	6.00	7.88	9.33	10.50	11.45	12.25	12.92
	6%	3.79	6.63	8.83	10.60	12.05	13.25	14.27
	7%	0.00	4.46	7.93	10.70	12.97	14.86	16.46
	8%	-7.71	0.00	6.00	10.80	14.73	18.00	20.77

Table 17: Fair P/E ratio based on different growth and ROE spread.

The table also points out an interesting property of P/E ratio. Under this simple model or a full-fledged DCF model, P/E ratio itself does not imply just growth. A P/E ratio of 10.8 could imply an ROE spread of 3% and a growth of 2% (high profitability and low growth), or an ROE spread of 0% and a growth 8% (low profitability and high growth).

The relation between growth and ROIC/ROE now proposes two issues with our established DCF model:

1. The growth in stable stage still comes with a cost that we cannot ignore.
2. The free cash flow number the terminal value based on does not reflect change in reinvestment rate and growth

For the discussion below we assume that ROIC/ROE stays constant in any stages. We also assume, without loss of generality, interest expense and taxes are 0 such that EBIT equals to NOPAT and it can be converted to FCFF directly through $1 - RR$.

Use FCF at year $N + 1$ In our established DCF model, our TV is based on FCF_N where N is length of growth stage. In the most simple two-stage DCF, FCF grows at a growth rate of g' at growth stage and $g = 2\%$ at stable stage. Recall that TV is defined in Equation 8, which is actually an infinite geometric series simplified into one term,

$$TV = FCF_N + FCF_N \cdot \frac{1+g}{1+r} + FCF_N \cdot \left(\frac{1+g}{1+r}\right)^2 + \dots$$

Problem arise when we use ROIC and RR to determine growth rate. With the introduction of RR, we also have to assume different RRs at growth and stable stage, which means that NOPAT at year $N + 1$ will grow at a different rate than FCFF, because FCFF is ultimately determined by reinvestment, not FCFF from the previous year. Let RR' be the growth stage reinvestment rate and RR be the perpetual reinvestment rate, and use ROIC in growth stage in place of ROIC,

$$\begin{aligned} \frac{NOPAT_N}{NOPAT_{N-1}} &= 1 + g' = 1 + ROIC \cdot RR' \\ \Rightarrow \frac{FCFF_N}{FCFF_{N-1}} &= \frac{NOPAT_N \cdot (1 - RR')}{NOPAT_{N-1} \cdot (1 - RR')} = 1 + g' \\ \frac{NOPAT_{N+1}}{NOPAT_N} &= 1 + g = 1 + ROIC \cdot RR \\ \Rightarrow \frac{FCFF_{N+1}}{FCFF_N} &= \frac{NOPAT_{N+1} \cdot (1 - RR)}{NOPAT_N \cdot (1 - RR')} = (1 + g) \cdot \frac{1 - RR}{1 - RR'} \end{aligned}$$

Therefore, TV must be based on FCF at year $N + 1$ instead of N , and FCF at year $N + 1$ must be recalculated according to stable RR and growth instead of extrapolated using RR' and g' .

By using FCF at year $N + 1$, we also have the term $1 - RR$ baked into our TV. With $RR = g/ROIC$, the requirement that ROIC is greater than WACC is still reflected in TV.

Rate of return spread In Table 16 we have seen that negative rate of return spread leads to lower P/E as growth goes up. This leads to the general notion that “rate of return spread must be positive” for a good investment since some sort of growth is still expected. If a company has negative rate of return spread, its value is destroyed as the company reinvests to grow, unless it decides to just grow at the bare minimum of 2%.

4 Stock Picking

Stock picking essentially is “buying good companies at a discount with wide margin of safety.”

1. How good a company is depends on its fundamental financial figures (margin, ROIC, etc)
2. How fairly a company is priced is revealed by DCF

Wide margin of safety is partly achieved by good fundamentals so that companies fare well even during downturns. It is also partly achieved by using conservative figures in valuation. As the saying goes, “it is better to be vaguely right than exactly wrong.”

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